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Operations Research: Duality

DUALITY

Primal – Dual

Every linear programming problem has associated with it another linear programming problem called the **dual.** The relationships between the dual problem and the original problem (called the **primal**) are extremely useful. When taking the dual of any LP, the given LP is referred to as the ***primal***. If the primal is a **max** problem, the dual will be a **min** problem and vice versa.

Knowledge of the dual provides interesting economic and sensitivity analysis insights. One of the key uses of duality theory lies in the interpretation and implementation of *sensitivity analysis.* Sensitivity analysis is a very important in linear programming study. Because most of the parameter values used in the original model are just *estimates* of future conditions, the effect on the optimal solution. Furthermore, certain parameter values (such as resource amounts) may represent *managerial decisions,* in which case the choice of the parameter values may be the main issue to be studied, which can be done through sensitivity analysis.

The dual of a ***normal max*** problem is a ***normal min*** problem.

Normal max problem is a problem in which all the variables are required to be nonnegative and all the constraints are ≤ constraints.

Normal min problem is a problem in which all the variables are required to be nonnegative and all the constraints are ≥ constraints.

Similarly, the dual of a normal min problem is a normal max problem.

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Finding the Dual of a Normal Max Problem

PRIMAL

$$\max z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$$

s.t.

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \leq b_2$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots$$

$$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \leq b_m$$

$$x_j \geq 0 \ (j = 1, 2, \ldots, n)$$

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$$\min w = b_1y_1 + b_2y_2 + \ldots + b_my_m$$

s.t.

$$a_{11}y_1 + a_{21}y_2 + \ldots + a_{m1}y_m \geq c_1$$

$$a_{12}y_1 + a_{22}y_2 + \ldots + a_{m2}y_m \geq c_2$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots$$

$$a_{1n}y_1 + a_{2n}y_2 + \ldots + a_{mn}y_m \geq c_n$$

$$y_i \geq 0 \ (i = 1, 2, \ldots, m)$$

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Finding the Dual of a Normal Min Problem

PRIMAL

min $w = b_1y_1 + b_2y_2 + \ldots + b_my_m$

s.t.

$$a_{11}y_1 + a_{21}y_2 + \ldots + a_{m1}y_m \geq c_1$$

$$a_{12}y_1 + a_{22}y_2 + \ldots + a_{m2}y_m \geq c_2$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots$$

$$a_{1n}y_1 + a_{2n}y_2 + \ldots + a_{mn}y_m \geq c_n$$

$$y_i \geq 0 \ (i = 1, 2, \ldots, m)$$

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max $z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$

s.t.

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \leq b_2$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots$$

$$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \leq b_m$$

$$x_j \geq 0 \ (j = 1, 2, \ldots, n)$$

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Rules for Constructing Dual Problem

Primal objective	Dual		
	Objective	Constraint	Variable sign
Maximization ≤	Minimization	≥	Unrestricted
Minimization ≥	Maximization	≤	Unrestricted

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Given our standard form for the *primal problem* at the left (perhaps after conversion from another form), its *dual problem* has the form shown to the right.

Primal Problem	*Dual Problem*
Maximize $Z = \sum_{j=1}^{n} c_j x_j,$	Minimize $W = \sum_{i=1}^{m} b_i y_i,$
subject to	subject to
$\sum_{j=1}^{n} a_{ij} x_j \le b_i,$ for $i = 1, 2, \ldots, m$	$\sum_{i=1}^{m} a_{ij} y_i \ge c_j,$ for $j = 1, 2, \ldots, n$
and	and
$x_j \ge 0,$ for $j = 1, 2, \ldots, n.$	$y_i \ge 0,$ for $i = 1, 2, \ldots, m.$

Thus, the dual problem uses exactly the same *parameters* as the primal problem, but in different locations. To highlight the comparison, now look at these same two problems in matrix notation (as introduced at the beginning of Sec. 5.2), where $\mathbf{c}$ and $\mathbf{y} = [y_1, y_2, \ldots, y_m]$ are row vectors but $\mathbf{b}$ and $\mathbf{x}$ are column vectors.

Primal Problem	*Dual Problem*
Maximize $Z = \mathbf{cx},$	Minimize $\mathbf{W} = \mathbf{yb},$
subject to	subject to
$\mathbf{Ax} \le \mathbf{b}$	$\mathbf{yA} \ge \mathbf{c}$
and	and
$\mathbf{x} \ge \mathbf{0}.$	$\mathbf{y} \ge \mathbf{0}.$

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Primal and dual problems for the Wyndor Glass Co. example

Primal Problem in Algebraic Form

Maximize $Z = 3x_1 + 5x_2,$

subject to

$$\begin{aligned} x_1 \qquad\quad &\le 4 \\ 2x_2 &\le 12 \\ 3x_1 + 2x_2 &\le 18 \end{aligned}$$

and $x_1 \ge 0, \quad x_2 \ge 0.$

Dual Problem in Algebraic Form

Minimize $W = 4y_1 + 12y_2 + 18y_3,$

subject to

$$\begin{aligned} y_1 \qquad + 3y_3 &\ge 3 \\ 2y_2 + 2y_3 &\ge 5 \end{aligned}$$

and

$y_1 \ge 0, \quad y_2 \ge 0, \quad y_3 \ge 0.$

Primal Problem in Matrix Form

Maximize $Z = [3, 5]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$

subject to

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \le \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$$

and

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \ge \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Dual Problem in Matrix Form

Minimize $W = [y_1, y_2, y_3]\begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix}$

subject to

$$[y_1, y_2, y_3]\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} \ge [3, 5]$$

and

$$[y_1, y_2, y_3] \ge [0, 0, 0].$$

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Weak duality property: If **x** is a feasible solution for the primal problem and **y** is a feasible solution for the dual problem, then

$\mathbf{cx} \leq \mathbf{yb}$.

For *any* pair of feasible solutions, this inequality must hold. The *maximum* feasible value of Z = **cx** (36) *equals* the *minimum* feasible value of the dual objective function W = **yb**.

Strong duality property: If $\mathbf{x}^*$ is an optimal solution for the primal problem and $\mathbf{y}^*$ is an optimal solution for the dual problem, then

$\mathbf{cx}^* = \mathbf{y}^*\mathbf{b}$.

These two properties imply that $\mathbf{cx} = \mathbf{yb}$ for feasible solutions if one or both of them are *not optimal* for their respective problems, whereas equality holds when both are optimal.

Duality theorem: The following are the only possible relationships between the primal and dual problems.

1. If one problem has *feasible solutions* and a *bounded* objective function (and so has an optimal solution), then so does the other problem, so both the weak and strong duality properties are applicable.

2. If one problem has *feasible solutions* and an *unbounded* objective function (and so *no optimal solution*), then the other problem has *no feasible solutions.*

3. If one problem has *no feasible solutions,* then the other problem has either *no feasible solutions* or an *unbounded* objective function.

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Finding the Dual of a Nonnormal Max Problem

- If the *i*th primal constraint is a ≥ constraint, the corresponding dual variable y_i must satisfy $y_i \leq 0$
- If the *i*th primal constraint is an equality constraint, the dual variable y_i is now unrestricted in sign (urs).
- If the *i*th primal variable is urs, the *i*th dual constraint will be an equality constraint

Finding the Dual of a Nonnormal Min Problem

- If the *i*th primal constraint is a ≤ constraint, the corresponding dual variable x_i must satisfy $x_i \leq 0$
- If the *i*th primal constraint is an equality constraint, the dual variable x_i is now urs.
- If the *i*th primal variable is urs, the *i*th dual constraint will be an equality constraint

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Economic Interpretation: When the primal is a normal max problem, the dual variables are related to the value of resources available to the decision maker. For this reason, dual variables are often referred to as *resource shadow prices*.

Example. PRIMAL

Let x_1, x_2, x_3 be the number of desks, tables and chairs produced. Let the weekly profit be $\$z$. Then, we must

$$\max z = 60x_1 + 30x_2 + 20x_3$$

$$\begin{aligned} \text{s.t.} \quad & 8x_1 + 6x_2 + x_3 \le 48 && \text{(Lumber constraint)} \\ & 4x_1 + 2x_2 + 1.5x_3 \le 20 && \text{(Finishing hour constraint)} \\ & 2x_1 + 1.5x_2 + 0.5x_3 \le 8 && \text{(Carpentry hour constraint)} \\ & x_1, x_2, x_3 \ge 0 \end{aligned}$$

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Suppose an entrepreneur wants to purchase all of Dakota's resources. In the dual problem y_1, y_2, y_3 are the resource prices (price paid for one board ft of lumber, one finishing hour, and one carpentry hour).
\$w is the cost of purchasing the resources. Resource prices must be set high enough to induce Dakota to sell. i.e. total purchasing cost equals total profit.

$$\min w = 48y_1 + 20y_2 + 8y_3$$

$$\begin{aligned} \text{s.t.} \quad & 8y_1 + 4y_2 + 2y_3 \ge 60 && \text{(Desk constraint)} \\ & 6y_1 + 2y_2 + 1.5y_3 \ge 30 && \text{(Table constraint)} \\ & y_1 + 1.5y_2 + 0.5y_3 \ge 20 && \text{(Chair constraint)} \\ & y_1, y_2, y_3 \ge 0 \end{aligned}$$

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TABLE Association between variables in primal and dual problems

	Primal Variable	Associated Dual Variable
Any problem	(Decision variable) x_j (Slack variable) x_{n+i}	$z_j - c_j$ (surplus variable) $j = 1, 2, \ldots, n$ y_i (decision variable) $i = 1, 2, \ldots, m$
Wyndor problem	Decision variables: x_1 x_2 Slack variables: x_3 x_4 x_5	$z_1 - c_1$ (surplus variables) $z_2 - c_2$ y_1 (decision variables) y_2 y_3

TABLE Corresponding primal-dual forms

Label	Primal Problem (or Dual Problem)		Dual Problem (or Primal Problem)
	Maximize Z (or W)		Minimize W (or Z)
	Constraint i:		Variable y_i (or x_i):
Sensible	$\leq$ form	$\longleftrightarrow$	$y_i \geq 0$
Odd	$=$ form	$\longleftrightarrow$	Unconstrained
Bizarre	$\geq$ form	$\longleftrightarrow$	$y_i' \leq 0$
	Variable x_j (or y_j):		Constraint j:
Sensible	$x_j \geq 0$	$\longleftrightarrow$	$\geq$ form
Odd	Unconstrained	$\longleftrightarrow$	$=$ form
Bizarre	$x_j' \leq 0$	$\longleftrightarrow$	$\leq$ form

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The primal-dual form for the radiation therapy example

Primal Problem		*Dual Problem*	
Minimize	$Z = 0.4x_1 + 0.5x_2$,	Maximize	$W = 2.7y_1' + 6y_2' + 6y_3$,
subject to		subject to	
(B)	$0.3x_1 + 0.1x_2 \le 2.7$ $\longleftrightarrow$	$y_1' \le 0$	(B)
(O)	$0.5x_1 + 0.5x_2 = 6$ $\longleftrightarrow$	y_2' unconstrained in sign	(O)
(S)	$0.6x_1 + 0.4x_2 \ge 6$ $\longleftrightarrow$	$y_3 \ge 0$	(S)
and		and	
(S)	$x_1 \ge 0$ $\longleftrightarrow$	$0.3y_1' + 0.5y_2' + 0.6y_3 \le 0.4$	(S)
(S)	$x_2 \ge 0$ $\longleftrightarrow$	$0.1y_1' + 0.5y_2' + 0.4y_3 \le 0.6$	(S)

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Example:

Primal	Dual
Max $Z=5x1+12x2+4x3$ St: $x1+2x2+x3 \leq 10$ $2x1-x2+3x3=8$ $X1,x2,x3 \geq 0$	Min $w=10y1+8y2$ St: $y1+2y2 \geq 5$ $2y1-y2 \geq 12$ $y1+3y2 \geq 4$ $y1 \geq 0$, $y2$ unrestricted

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One primal-dual form for the radiation therapy example

Primal Problem		*Dual Problem*	
Maximize	$-Z = -0.4x_1 - 0.5x_2,$	Minimize	$W = 2.7y_1 + 6y_2 + 6y_3',$
subject to		subject to	
(S)	$0.3x_1 + 0.1x_2 \le 2.7$ $\longleftrightarrow$	$y_1 \ge 0$	(S)
(O)	$0.5x_1 + 0.5x_2 = 6$ $\longleftrightarrow$	y_2 unconstrained in sign	(O)
(B)	$0.6x_1 + 0.4x_2 \ge 6$ $\longleftrightarrow$	$y_3' \le 0$	(B)
and		and	
(S)	$x_1 \ge 0$ $\longleftrightarrow$	$0.3y_1 + 0.5y_2 + 0.6y_3' \ge -0.4$	(S)
(S)	$x_2 \ge 0$ $\longleftrightarrow$	$0.1y_1 + 0.5y_2 + 0.4y_3' \ge -0.5$	(S)